

ME 204 A : MECHANICS OF SOLIDS

2L, 1T 3 Hours, 100 Marks

Direct Stresses in Machine Parts : Analysis of stress and strain. -Hooke's law, stress-strain diagram, Poisson's ratio, Modulus of elasticity, Relation between elastic constants

Simple stresses : Tensile, compressive, shear, crushing and bearings. criteria. Hoop stress in thin cylindrical and spherical vessels subjected to internal pressure

Machine Members in Bending : Theory of bending of straight and curved beams. Bending and shear stress distribution in beams. Shear force and bending moment diagrams. Moment of inertia of different machine element sections

Deflection and stiffness : Mecauly's method, area moment method, numerical integration method and energy method Numerical problems of line shafts under simple loading with pulleys, ears, flywheel etc. Brief introduction to analysis of shafts with three Supports.

Machine Members in Torsion : Introduction to torsion of circular solid and hollow sections. Angular deformation and strain energy.

Stress in Slender Machines Members under axial Compressive Loads : Analysis of long and short columns. Euler's theory and effect of end conditions on stresses. Emperical formulas

Machine Members subjected to Combined Loading: Combination of axial, bending, and torsional stresses. Concept of principal stresses and planes. Mohr's circle of stresses. 2D state of strain, Plane stress and Plane strain conditions. Mohr's circle for strain. Theories of failure. Simple problems of line shaft.

SN Name of Authors /Books /Publisher

Year of

Pub.

1 Timoshenko, S.P., and Gere, J.M., “Mechanics of Materials”, 2nd Ed., CBS Publishers

2 Crandall, S.H., Dahl, N.C., and Lardner, T.J., “An Introduction to the Mechanics of Solids”, TMGH

3 Strength of materials , Ramamurtham

4 Punmia, Jain and Jain, “Mechanics of Materials”, Laxmi Publication

5 Popov, E.P., Nagarajan, S., and Lu, Z. A., “Mechanics of Materials”, Prentice-Hall of India